



MODULE NAME: 5YYD0010 Quantitative Methods

MODULE CONVENOR: Dr. Yundan Gong

MODULE ASSESSMENT PATTERN (Semester 1):

Assessment	Duration/Length	% of final grade	Deadline
CW2 – 1,500 Word Essay	1,500 words	50	Wednesday, 29 April 2025 by 15:

ASSESSMENT QUESTIONS AND INSTRUCTIONS

Your essay should include:

- (1) descriptive statistics and data visualisation;
- (2) hypothesis development based on relevant literature;
- (3) regression model selection and OLS assumption discussions based ON relevant literature
- (4) regression results with appropriate controls;
- (5) result interpretation with reference to relevant literature and policy insights.

Giving this is an essay, please don't forget introduction and conclusion. Reference is also required.

The word limitation is 1,500 word excluding table, figure, reference and appendix.

Data Source: World Development Indicators (WDI), World Bank, 2000-2019.

The cross-section file (5YYD0010_CrossSection.dta) contains country averages over 2000-2019 and is suitable for basic data visualisation and OLS regression analysis.

You are encouraged to extend the classic papers by exploring additional variables in the dataset (e.g., different types of trade, FDI, human capital). Consider how contemporary issues (such as the US-China trade war, climate policy, and global supply chain disruptions) might affect the relationships you examine. The interaction terms (e.g., trade_post2015, tariff_post2018) are pre-calculated and can be used directly in your regressions to explore whether relationships changed after specific policy events.

For advanced analysis: If you would like to use the panel data, please discuss with your lecturer and tutor for the option. The panel data is available upon to request.



Please answer ONE of the following questions.

Research Questions:

Question 1: Trade and Environment

Does international trade increase or decrease CO2 emissions? Do the 'gains from trade' (e.g., technology transfer, stricter regulations) outweigh the environmental costs, or does trade facilitate pollution?

Dataset: 5YYD0010_CrossSection.dta

Suggested Reading

Frankel, J. & Rose, A. (2005). 'Is Trade Good or Bad for the Environment? Sorting Out the Causality.' *Review of Economics and Statistics*, 87(1), 85-91.

Theoretical background and model

Frankel and Rose (2005) identify competing hypotheses you may use: (1) Race-to-the-bottom: Open economies adopt looser environmental standards to attract investment, increasing pollution. (2) Gains-from-trade: Trade brings cleaner technology, economies of scale, and higher income that enables stricter regulations. (3) Pollution Haven Hypothesis: Developing countries become 'pollution havens' as dirty industries relocate from rich countries. (4) Environmental Kuznets Curve: Pollution first rises then falls with income (inverted-U relationship).

Question 2: Tariffs and Trade

Do tariff barriers reduce international trade? How did the US-China trade war (post-2018) affect this relationship, particularly for East Asian economies?

Dataset: 5YYD0010_CrossSection.dta (same dataset, includes tariff variables)

Suggested Reading



Rose, A. (2004). 'Do We Really Know That the WTO Increases Trade?' *American Economic Review*, 94(1), 98-114.

Baier, S. & Bergstrand, J. (2007). 'Do Free Trade Agreements Actually Increase Members' International Trade?' *Journal of International Economics*, 71(1), 72-95.

Theoretical background and model

Standard trade theory predicts that tariffs increase the cost of trade and therefore reduce international trade. Your task is to test whether the empirical evidence supports this prediction. The suggested readings examine how trade agreements and institutions affect trade flows. You can extend their approach to analyse the impact of the US-China trade war (post-2018) on global trade patterns, particularly for East Asian economies that are deeply integrated into global supply chains.